Práctica dos: Sistema cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



Nombre del alumno: **Tessa Rincón Jiménez**

Número de control: **22210795**

Correo institucional: **l22210795@tectijuana.edu.mx**

Asignatura: **Modelado de Sistemas Fisiológicos**

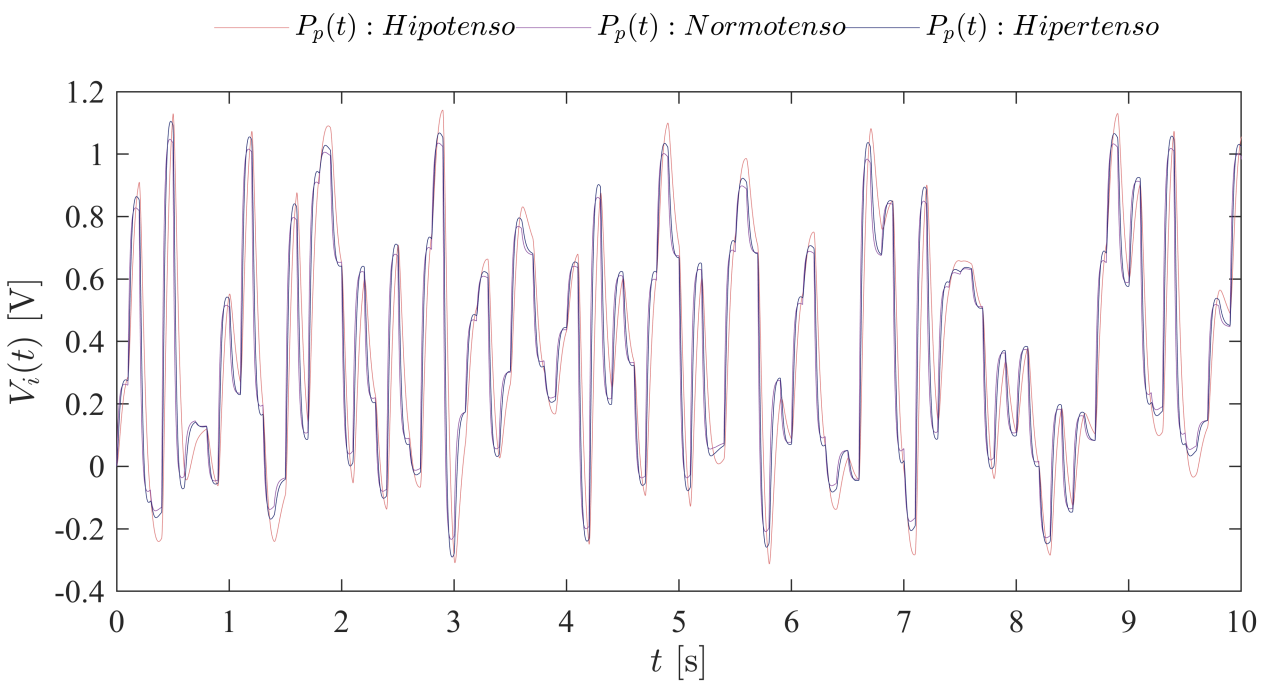
Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos de la simulación

```
clc; clear; close all; warning('off', 'all')
open_system('P3D')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
x = sim('P3D', parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(x.Pa, 'signal.xlsx');
```

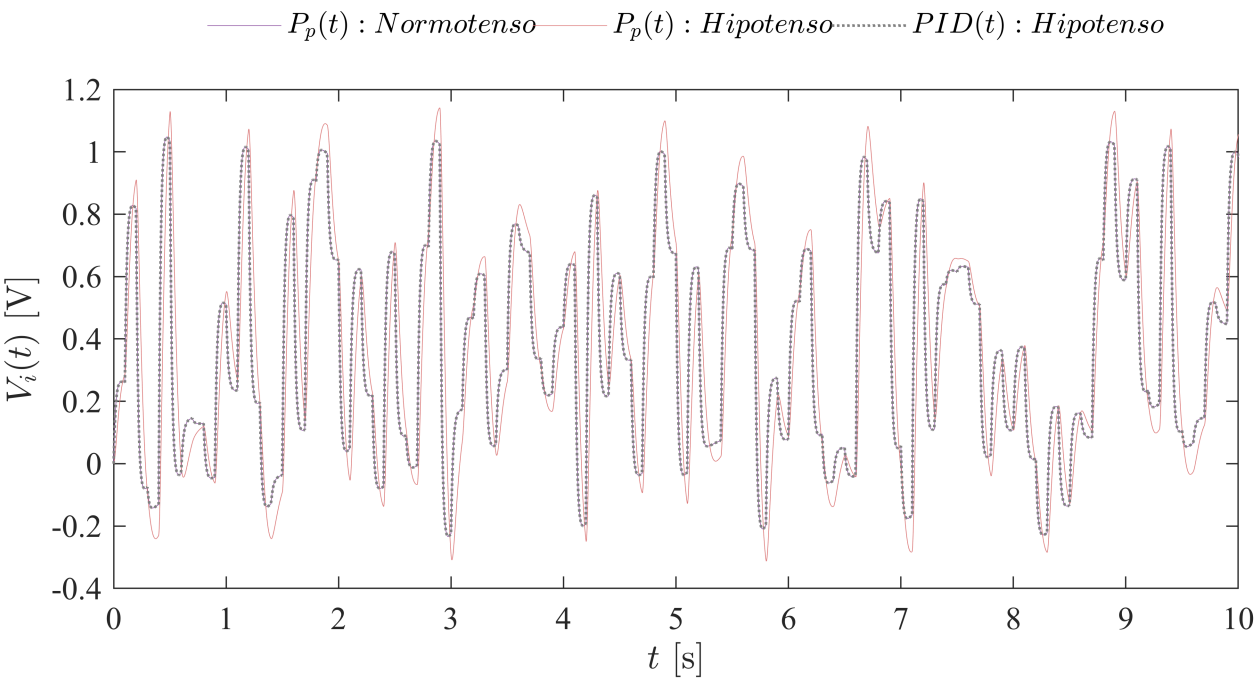
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t, x.PpL, x.PpN, x.PpH, 'individuos')
```



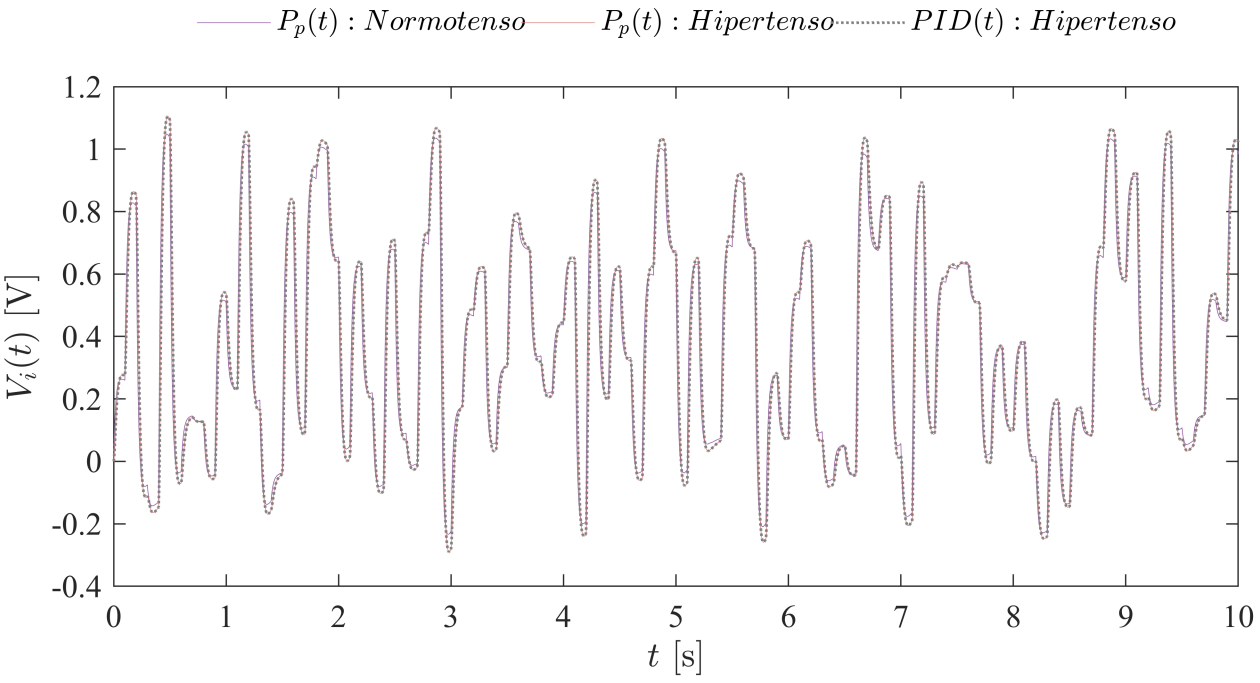
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t, x.PpN, x.PpL, x.PIDL, 'hipotenso')
```



Respuestas: Normotenso, Hipertenso y Controlador

```
plotsignals(x.t, x.PpN, x.PpH, x.PIDH, 'hipertenso')
```

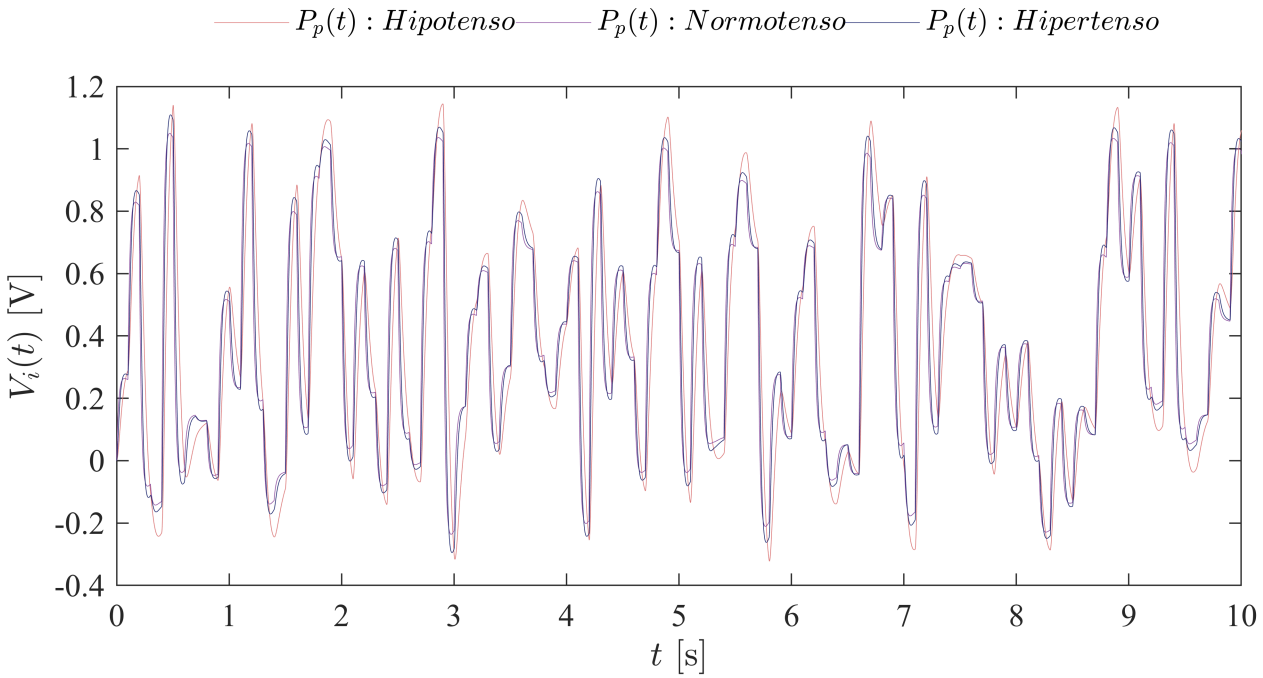


Modelo de EDOs

```
par_hipo = [0.01, 0.001, 1.00, 3.50]; %[Z,L,R,C]
par_normo = [0.01, 0.001, 1.00, 1.500];
```

```
par_hiper = [0.01, 0.001, 1.00, 1.000];

PpN = cardio(par_normo, Pa, t);
PpL = cardio(par_hipo, PpN, t);
PpH = cardio(par_hiper, PpN, t);
plotsignals(t, PpL, PpN, PpH, 'odes')
```




```
%% Función cardio
```

```
function Pp = cardio(par, Pa, t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    Fl = zeros(length(t),1); Fl(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1:n
        Fl(i+1) = Fl(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (Fl(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end

%% Función plotsignals
function plotsignals(t, x1, x2, x3, signal)
    set(figure(), 'color', 'w')
    set(gcf, 'units', 'inches', 'position', [1, 1, 10, 5])
    set(gca, 'fontname', 'times new roman', 'fontsize', 12)
    hold on; grid off; box on;
function Pp = cardio(par, Pa, t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
```

```
    n = round(t(end)/dt);
    Fl = zeros(length(t),1); Fl(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1:n
        Fl(i+1) = Fl(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (Fl(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end
```

Funcion: Respuestas a la simulación

```
   clr1 = [220, 120, 120]/255;
    clr2 = [140, 80, 160]/255;
    clr3 = [30, 40, 110]/255;
    clr4 = [0.5, 0.5, 0.5];

    order = [clr1; clr2; clr3; clr4];
    colororder(order)

    if signal == "individuos"
        plot(t, x1, '-', 'LineWidth', 0.25, 'Color', clr1) % Hipotenso
        plot(t, x2, '-', 'LineWidth', 0.25, 'Color', clr2) % Normotenso
        plot(t, x3, '-', 'LineWidth', 0.25, 'Color', clr3) % Hipertenso
        L = legend('$P_p(t):Hipotenso$', '$P_p(t):Normotenso$',
'$P_p(t):Hipertenso$');
    elseif signal == "hipotenso"
        plot(t, x1, '-', 'LineWidth', 0.25, 'Color', clr2) % Normotenso
        plot(t, x2, '-', 'LineWidth', 0.25, 'Color', clr1) % Hipotenso
        plot(t, x3, ':', 'LineWidth', 1, 'Color', clr4)    % Controlador Hipotenso
        L = legend('$P_p(t):Normotenso$', '$P_p(t):Hipotenso$',
'$PID(t):Hipotenso$');
    elseif signal == "hipertenso"
        plot(t, x1, '-', 'LineWidth', 0.25, 'Color', clr2) % Normotenso
        plot(t, x2, '-', 'LineWidth', 0.25, 'Color', clr1) % Hipotenso
        plot(t, x3, ':', 'LineWidth', 1, 'Color', clr4)    % Controlador Hipertenso
        L = legend('$P_p(t):Normotenso$', '$P_p(t):Hipertenso$',
'$PID(t):Hipertenso$');
    elseif signal == "odes"
        plot(t, x1, '-', 'LineWidth', 0.25, 'Color', clr1) % Hipotenso
        plot(t, x2, '-', 'LineWidth', 0.25, 'Color', clr2) % Normotenso
        plot(t, x3, '-', 'LineWidth', 0.25, 'Color', clr3) % Hipertenso
        L = legend('$P_p(t):Hipotenso$', '$P_p(t):Normotenso$',
'$P_p(t):Hipertenso$');
    end

    set(L, 'interpreter', 'latex', 'location', 'northoutside', 'orientation',
'horizontal', 'box', 'off')
    xlabel('$t$ [s]', 'interpreter', 'latex')
    ylabel('$V_i(t)$ [V]', 'interpreter', 'latex')
    xlim([0 10]); xticks(0:1:10)
```

```
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
    exportgraphics(gcf, [signal, 'matlab.pdf'], 'ContentType', 'vector')
end
```